

Sreeram Melpadi

Software Engineer

📍 Buffalo, NY

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🌐 [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Professional Summary

Master's student in Computer Science at University at Buffalo bound to graduate in Dec 2025. 3.5 years experience as a Software Engineer in full-stack development, distributed systems, and large-scale deployments. Eligible for F-1 OPT/STEM OPT (36 Mo).

Education

University at Buffalo, Buffalo, NY, USA

Aug 2024 - Dec 2025

M.S. in Computer Science (AI/ML Specialization)

GPA: 4.0 / 4.0

The National Institute of Engineering, Mysuru, KA, India

Aug 2017 - June 2021

B.E. in Electronics and Communication Engineering

GPA: 8.73 / 10

Work Experience

Sandvine (now AppLogic Networks), Bangalore

Senior Software Engineer (R&D)

Dec 2023 - July 2024

- Worked **on-site** in Australia for 40 days as the team's **SME**, leading a production deployment at **NBN Co**, Australia's largest network provider, ensuring better network QoE for **hundreds of thousands of subscribers** and retained a \$1M deal.
- **Led** the design, architecture, and development of an end-end Web app with **Django**(Backend), **React/TypeScript** (Frontend), and **Vertica**(DB) to visualize subscriber and network locations' QoE, providing real-time insights for network performance.

Software Development Engineer (R&D)

July 2021 - Nov 2023

- Performed **bulk data analysis** on millions of subscribers' network packet meta-data like bandwidth, round-trip time, and packet loss to understand, analyze, and find potential areas of QoE improvement. (Python, Pandas, Numpy)
- Enhanced the disaster recovery by implementing Active-Active High Availability and Active-Passive Geo Redundancy by **containerizing** and clustering servers while **load-balancing** across the data centers with an **RTO of 4 minutes**. (Kubernetes)
- Designed and integrated efficient asynchronous **gRPC** APIs for **microservices-based** distributed systems, enabling horizontal scaling of processing nodes thereby increasing request handling rate by **64%**. (Go, Linux, gRPC)
- Optimized throughput performance of Congestion Management product by **850%** using efficient DSA and design and reducing redundancy over the product pipeline after profiling and analyzing expensive method calls using **cProfile** and **py-spy**.

Projects

Distributed Key-Value Store using Consistent Hashing and gRPC ([Link](#))

Oct, 2025

- Developing a distributed Key-Value store in **Go** with sharded Containerized nodes managed by an external coordinator leveraging **consistent hashing** for key partitioning, **replication**, and fault tolerance, with **gRPC/Protobuf** for inter-service communication. Going ahead, planning to add auto-scaling, health checks, and dynamic rebalancing.

TheraPT: Empathetic Dialogue Generation using Transformers With CoT Reasoning ([Link](#))

Apr, 2025

- Developed a mental health-focused **conversational agent** by fine-tuning **GPT-2** on counseling datasets along with **BART**-based context compression to summarize prior user interactions and enable long-range coherence. Integrated **chain-of-thought** prompting to induce reasoning. Achieved strong semantic alignment with human responses (**BERTScore F1: 0.8948**).

Face Mask Detection, Recognition, and Attendance System ([Link](#))

May, 2021

- Developed a face mask detection and (face) recognition system using live-feed from **Raspberry Pi**, marking attendance with mask status in a remote database. Trained a **CNN** model on **GPU** with 89% accuracy using **TensorFlow/Keras**, scraped images with **Beautiful Soup**, and OpenCV for human detection. Deployed the system on **Cloud** using **AWS EC2, RDS (MySQL)** and **Flask**.

Skills

Programming Languages: Python, Go, Typescript, C++, SQL, Bash

Frameworks & Libraries: ReactJS, Django, Flask, REST, gRPC, PyTorch, Pandas, Apache Spark

Cloud & Infrastructure: AWS (EC2, RDS), Docker, Kubernetes, Consul, Kafka, Redis, Linux, Git, CI/CD

Databases & Tools: Vertica, PostgreSQL, MongoDB, TCP/IP, Excel, Tableau, Kaggle

Soft Skills & Methodologies: Quick Learning, Problem-Solving, Customer Focused, Communication, Teamwork, Agile (Scrum)